

Title

Soft Robotic Exoskeleton Training Enhances Gait Parameters in Parkinson Disease: Results of a Pilot Randomized Study

Abstract

Robotic rehabilitation tools may provide advantages over conventional therapy by delivering precisely-calibrated motor assistance. Fifty-six individuals with Parkinson disease underwent randomization to standard rehabilitation versus standard rehabilitation plus four weeks of soft exoskeleton training (five times weekly, 20 minutes per session). The exoskeleton-augmented group demonstrated significantly greater improvements in stride length symmetry and walking speed compared to controls. Specifically, stride length increased by approximately 0.15 meters on both sides, and ankle plantarflexion flexibility improved substantially. Measures of motor severity on standardized clinical scales decreased more in the exoskeleton group. Additionally, participants receiving robotic assistance showed improved balance-related subscale scores and modest gains in cognitive screening measures. Improvements in ankle mobility showed strong positive correlation with overall gait velocity gains, suggesting a mechanistic link. No serious adverse events occurred. Soft robotic exoskeleton training represents a feasible and effective adjunct to conventional rehabilitation for reducing gait dysfunction in Parkinson disease.